and that's what counts. That's the sense in which the speed limit is enforced.

In my science interpretation of the movie, since the planet always keeps the same face pointed toward Gargantua (Figure 17.4), it must spin at the same rate as it orbits, ten revolutions per second. How can it possibly spin so fast? Won't centrifugal forces tear it apart? No; and again the savior is the whirl of space. The planet would feel *no* disruptive centrifugal forces if it were spinning at precisely the same rate as space near it whirls, which it is almost doing! So centrifugal forces due to its rotation, in fact, are weak. If, instead, it were nonspinning relative to the distant stars, it would turn at ten revolutions per second relative to whirling space and so would be torn apart by centrifugal forces. It's weird what relativity can do.

Giant Waves on Miller's Planet

What could possibly produce the two gigantic water waves, 1.2 kilometers high, that bear down on the Ranger as it rests on Miller's planet (Figure 17.5)?

I searched for a while, did various calculations with the laws of physics, and found two possible answers for my science interpretation of the movie. Both answers require that the planet be *not quite* locked to Gargantua. Instead it must rock back and forth relative to Gargantua by a small amount, from the orientation on the left of Figure 17.6, to that on the right, then back to the left, and so on.

This rocking is a natural thing, as you can see by looking at Gargantua's tidal gravity.

In Figure 17.6, I describe the tidal gravity by *tendex lines* (Chapter 4). No matter which way the planet is tilted (left or right in Figure 17.6), Gargantua's blue squeezing tendex lines push its sides in, which drives